(a) Path Solver

Modal Weighting

$$w_{ij,m} = \frac{\sqrt{T_m}\, e^{-\alpha_m l_{2,m}}}{l_1 + l_{2,m} + l_3}$$

Detector

Surface Gradient ∇s

l_3

(q_{in}, s_{in})

$l_{2,m}$

(q_{out}, s_{out})

Skull

l_1

$$q^{k+1}_{in/out} = q^{k}_{in/out} - \eta_r \nabla_{q_{in/out}} \tau$$

Source

Initial Path ⟶ Stationary Path

(b) Transcranial Propagation

Coupling Water $Z_{w'}$

Detector

Trans. Coeff. $\tilde{T}_{out} = \frac{2Z'_{out,\omega}}{Z'_{out,\omega} + Z'_{out,m}}$

S Mode

P Mode

Skull $Z_{m'}$

$\tilde{T}_{in} = \frac{2Z'_{in,m}}{Z'_{in,\omega} + Z'_{in,m}}$

Source

Intracranial Fluid $Z_{w'}$

(c) Forward Model

$\tau_p \omega_p$

$\tau_s \omega_s$

Gaussian Source

$$p(t) = \sum_{m=P,S} \omega_m h(t - \tau_m)$$

Modal delay & weight

Sparse temporal scatter

Gaussian Convolution

(d) Iterative Reconstruction

Source x

Forward $\mathcal{A}$

Predicted Signal $\mathcal{A}x$

Residual $\mathcal{A}x - \mathrm{b}$

Regularzation

Hessian

TV

L1

Adjoint $\mathcal{A}^*$

Gradient Update